AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.072 / 0.0158734
Time	2026-07-20 20:00 UTC
Location	29.7644, -95.0781
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 718	29.01 to 100; mean 71.3193	46.79 at 2026-07-20 19:55Z; dt 5 min
asos	temperature	degC	7 / 718	23.1111 to 37; mean 29.7344	34 at 2026-07-20 19:55Z; dt 5 min
asos	wind_direction	degree	7 / 872	0 to 360; mean 183.268	0 at 2026-07-20 19:55Z; dt 5 min
asos	wind_speed	m/s	7 / 920	0 to 7.20222; mean 2.4246	0 at 2026-07-20 19:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5915.17 to 5936.85; mean 5925.97	5927.31 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	8 / 96	17.7446 to 1614.77; mean 632.794	1291.56 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	8 / 96	29.7056 to 46.5607; mean 36.2151	33.2762 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	8 / 96	1005.31 to 1015.78; mean 1010.7	1013.17 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	t_850	degC	8 / 96	19.7215 to 23.3589; mean 22.0754	22.1796 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	8 / 96	-2.78129 to 3.37582; mean 1.3262	1.11533 at 2026-07-20 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	8 / 96	-1.86296 to 4.7729; mean 1.1929	0.10071 at 2026-07-20 18:00Z; dt 120 min
openaq	ozone	ppm	16 / 1038	-0.004 to 0.094; mean 0.0208	0.072 at 2026-07-20 20:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 142	29 to 68; mean 44.2887	61 at 2026-07-20 20:00Z; dt 0 min
openaq	pm25	ug/m3	73 / 2812	0.3 to 28.8; mean 7.4711	19.7 at 2026-07-20 20:00Z; dt 0 min
openweather	cloud_cover	percent	4 / 284	0 to 100; mean 21.8239	6 at 2026-07-20 20:01Z; dt 1.9 min
openweather	humidity	percent	4 / 284	40 to 96; mean 73.1549	57 at 2026-07-20 20:01Z; dt 1.9 min
openweather	precipitation	mm/h	4 / 284	0 to 0; mean 0	0 at 2026-07-20 20:01Z; dt 1.9 min
openweather	pressure	hPa	4 / 284	1009 to 1017; mean 1013.07	1013 at 2026-07-20 20:01Z; dt 1.9 min
openweather	temperature	degC	4 / 284	24.14 to 37.79; mean 30.5224	35.89 at 2026-07-20 20:01Z; dt 1.9 min
openweather	wind_direction	degree	4 / 284	0 to 343; mean 215.486	184 at 2026-07-20 20:01Z; dt 1.9 min
openweather	wind_speed	m/s	4 / 284	0.4 to 5.9; mean 2.1399	1.63 at 2026-07-20 20:01Z; dt 1.9 min
purpleair	pm25	ug/m3	63 / 4511	0 to 149.3; mean 9.1181	7.7 at 2026-07-20 20:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0254212 to 0.0265722; mean 0.0261	0.0262849 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_co_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000176599 to 0.000224886; mean 0.0002	0.000213688 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_no2_column	mol/m^2	3 / 3	3.22877e-05 to 5.87309e-05; mean 0	5.83485e-05 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_no2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-8.60324e-06 to 4.32022e-05; mean 0	-8.60324e-06 at 2026-07-20 19:49Z; dt 10.4 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-20 19:49Z; dt 10.4 min
tceq	co	ppm	3 / 185	0 to 0.7; mean 0.1784	0.1 at 2026-07-20 20:00Z; dt 0 min
tceq	no2	ppb	12 / 683	-2.5 to 44.3; mean 5.1303	11.5 at 2026-07-20 20:00Z; dt 0 min
tceq	so2	ppb	3 / 181	-0.6 to 6.9; mean 0.1122	1.7 at 2026-07-20 20:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-19 06Z 93.85, 12Z 74.29, 18Z 53.98, 07-20 00Z 79.43, 06Z 93.45, 12Z 74.98, 18Z 48.76, 07-21 00Z 75.65, 06Z 92.16, 12Z 73.32, 18Z 39.33, 07-22 00Z 61.22, 06Z 76.19
asos	temperature	07-19 06Z 25.17, 12Z 29.51, 18Z 34, 07-20 00Z 28.82, 06Z 25.33, 12Z 29.14, 18Z 34.4, 07-21 00Z 29.33, 06Z 25.29, 12Z 29.09, 18Z 35.47, 07-22 00Z 30.65, 06Z 27.62
asos	wind_direction	07-19 06Z 62.71, 12Z 190.3, 18Z 139.6, 07-20 00Z 162.1, 06Z 145.7, 12Z 226, 18Z 159.2, 07-21 00Z 178.9, 06Z 199.3, 12Z 261.2, 18Z 229.4, 07-22 00Z 186.8, 06Z 248.8
asos	wind_speed	07-19 06Z 0.6859, 12Z 2.417, 18Z 2.209, 07-20 00Z 3.021, 06Z 1.75, 12Z 2.545, 18Z 2.209, 07-21 00Z 2.984, 06Z 2.261, 12Z 3.595, 18Z 2.414, 07-22 00Z 2.254, 06Z 2.901
noaa_gfs	gh_500	07-19 12Z 5923, 18Z 5926, 07-20 00Z 5922, 06Z 5930, 12Z 5920, 18Z 5928, 07-21 00Z 5923, 06Z 5926, 12Z 5917, 18Z 5932, 07-22 00Z 5930, 06Z 5935
noaa_gfs	pbl_height	07-19 12Z 180.6, 18Z 1341, 07-20 00Z 798.1, 06Z 247.3, 12Z 149.4, 18Z 1284, 07-21 00Z 856.6, 06Z 224.3, 12Z 222.4, 18Z 1242, 07-22 00Z 923.8, 06Z 123.4
noaa_gfs	precipitable_water	07-19 12Z 34.75, 18Z 36.26, 07-20 00Z 39.58, 06Z 34.69, 12Z 30.31, 18Z 32.87, 07-21 00Z 36.08, 06Z 34.18, 12Z 32.38, 18Z 36.59, 07-22 00Z 41.33, 06Z 45.56
noaa_gfs	surface_pressure	07-19 12Z 1014, 18Z 1013, 07-20 00Z 1011, 06Z 1012, 12Z 1012, 18Z 1012, 07-21 00Z 1009, 06Z 1009, 12Z 1011, 18Z 1010, 07-22 00Z 1008, 06Z 1008
noaa_gfs	t_850	07-19 12Z 21.94, 18Z 21.4, 07-20 00Z 20.94, 06Z 22.66, 12Z 22.62, 18Z 22.26, 07-21 00Z 21.38, 06Z 23.02, 12Z 22.31, 18Z 22.15, 07-22 00Z 21.75, 06Z 22.47
noaa_gfs	u_10m	07-19 12Z 0.7484, 18Z 1.356, 07-20 00Z -0.8788, 06Z 1.619, 12Z 1.532, 18Z 1.818, 07-21 00Z -0.3229, 06Z 1.877, 12Z 2.705, 18Z 2.268, 07-22 00Z 0.9234, 06Z 2.268

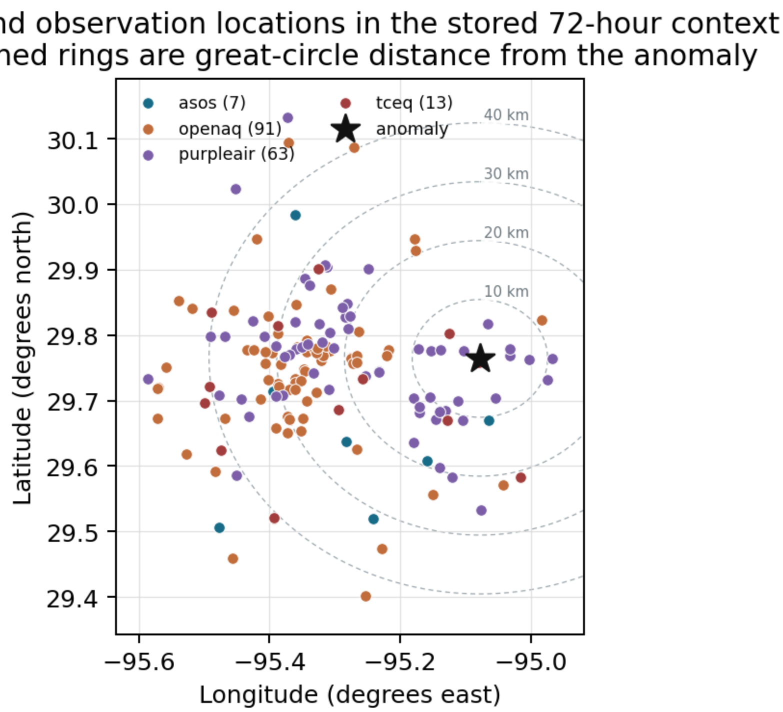
Source	Metric	Values
noaa_gfs	v_10m	07-19 12Z 1.512, 18Z 1.434, 07-20 00Z 2.877, 06Z 2.366, 12Z 0.4644, 18Z 0.5732, 07-21 00Z 2.195, 06Z 2.3, 12Z 0.2966, 18Z -0.7307, 07-22 00Z -0.1292, 06Z 1.158
openaq	ozone	07-19 06Z 0.007859, 12Z 0.01438, 18Z 0.03453, 07-20 00Z 0.01715, 06Z 0.006741, 12Z 0.01062, 18Z 0.04376, 07-21 00Z 0.01854, 06Z 0.006247, 12Z 0.01129, 18Z 0.04487, 07-22 00Z 0.02856, 06Z 0.007105
openaq	pm10	07-19 06Z 35.25, 12Z 35.75, 18Z 46.83, 07-20 00Z 45.25, 06Z 44.75, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.55, 06Z 39.83, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 44
openaq	pm25	07-19 06Z 8.839, 12Z 6.654, 18Z 8.287, 07-20 00Z 7.733, 06Z 8.757, 12Z 8.054, 18Z 6.761, 07-21 00Z 6.957, 06Z 7.531, 12Z 7.138, 18Z 5.843, 07-22 00Z 6.319, 06Z 6.397
openweather	cloud_cover	07-19 06Z 90.56, 12Z 92.95, 18Z 81.62, 07-20 00Z 12.04, 06Z 5.64, 12Z 2.958, 18Z 3.208, 07-21 00Z 3.739, 06Z 0.32, 12Z 0.5217, 18Z 1.64, 07-22 00Z 3.583, 06Z 5.571
openweather	humidity	07-19 06Z 91.69, 12Z 81.62, 18Z 60.38, 07-20 00Z 75.61, 06Z 90.72, 12Z 80.17, 18Z 53.25, 07-21 00Z 72.3, 06Z 89.2, 12Z 79.52, 18Z 47.8, 07-22 00Z 61.62, 06Z 77.29
openweather	precipitation	07-19 06Z 0, 12Z 0, 18Z 0, 07-20 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0
openweather	pressure	07-19 06Z 1016, 12Z 1017, 18Z 1014, 07-20 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010
openweather	temperature	07-19 06Z 25.79, 12Z 28.92, 18Z 35.34, 07-20 00Z 30.51, 06Z 25.94, 12Z 28.87, 18Z 35.82, 07-21 00Z 30.83, 06Z 25.92, 12Z 28.84, 18Z 36.52, 07-22 00Z 31.93, 06Z 28.22
openweather	wind_direction	07-19 06Z 188.7, 12Z 212.5, 18Z 186.3, 07-20 00Z 181.8, 06Z 186.9, 12Z 256, 18Z 221.3, 07-21 00Z 184.5, 06Z 196.4, 12Z 272.9, 18Z 268.1, 07-22 00Z 232.5, 06Z 174.7
openweather	wind_speed	07-19 06Z 1.699, 12Z 1.75, 18Z 2.181, 07-20 00Z 2.799, 06Z 1.84, 12Z 2.049, 18Z 1.876, 07-21 00Z 2.613, 06Z 2.282, 12Z 2.284, 18Z 1.649, 07-22 00Z 2.371, 06Z 2.72
purpleair	pm25	07-19 06Z 10.72, 12Z 8.548, 18Z 8.768, 07-20 00Z 9.176, 06Z 10.87, 12Z 9.812, 18Z 7.692, 07-21 00Z 8.914, 06Z 9.647, 12Z 8.489, 18Z 7.863, 07-22 00Z 9.88, 06Z 7.94
sentinel5p	s5p_aer_ai_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_ch4_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_co_column	07-19 18Z 0.02551, 07-20 18Z 0.02628, 07-21 18Z 0.02656
sentinel5p	s5p_co_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_hcho_column	07-19 18Z 0.0001766, 07-20 18Z 0.000213, 07-21 18Z 0.0002243
sentinel5p	s5p_hcho_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_no2_column	07-19 18Z 3.229e-05, 07-20 18Z 5.835e-05, 07-21 18Z 5.873e-05
sentinel5p	s5p_no2_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_o3_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
sentinel5p	s5p_so2_column	07-19 18Z 2.991e-06, 07-20 18Z 1.73e-05, 07-21 18Z 3.63e-05
sentinel5p	s5p_so2_granule_available	07-19 18Z 1, 07-20 18Z 1, 07-21 18Z 1
tceq	co	07-19 06Z 0.2, 12Z 0.2111, 18Z 0.2, 07-20 00Z 0.2667, 06Z 0.2056, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222, 06Z 0.1222, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.08889
tceq	no2	07-19 06Z 3.488, 12Z 2.541, 18Z 3.297, 07-20 00Z 3.545, 06Z 4.577, 12Z 6.21, 18Z 6.136, 07-21 00Z 6.723, 06Z 5.171, 12Z 5.402, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.025
tceq	so2	07-19 06Z -0.5222, 12Z 0.1056, 18Z 0.4389, 07-20 00Z -0.2875, 06Z -0.3471, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213, 06Z -0.2778, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2778

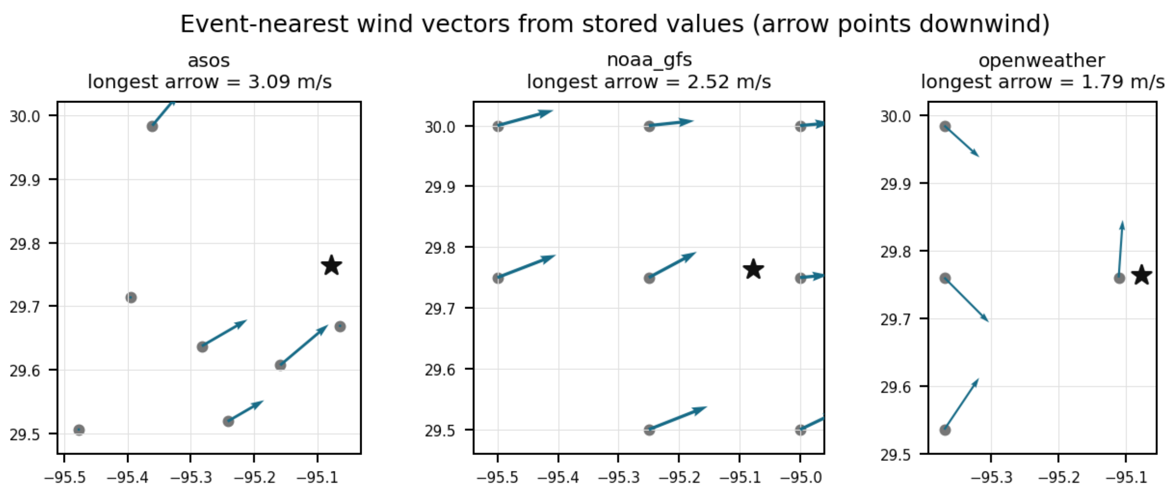
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.672 km) 73.13, EFD (19.125 km) 70.51, HOU (24.259 km) 71.55, MCJ (31.106 km) 67.98, LVJ (31.547 km) 73.4, IAH (36.618 km) 70.62, AXH (48.068 km) 72.33
asos	temperature	T41 (10.672 km) 29.73, EFD (19.125 km) 30.23, HOU (24.259 km) 29.82, MCJ (31.106 km) 30.25, LVJ (31.547 km) 29.26, IAH (36.618 km) 30.02, AXH (48.068 km) 29.07
asos	wind_direction	T41 (10.672 km) 198.3, EFD (19.125 km) 189.7, HOU (24.259 km) 218.2, MCJ (31.106 km) 205.5, LVJ (31.547 km) 163.9, IAH (36.618 km) 189.4, AXH (48.068 km) 128.7
asos	wind_speed	T41 (10.672 km) 3.011, EFD (19.125 km) 3.261, HOU (24.259 km) 3.119, MCJ (31.106 km) 3.011, LVJ (31.547 km) 1.74, IAH (36.618 km) 2.254, AXH (48.068 km) 1.051
noaa_gfs	gh_500	gfs:29.75,-95.00 (7.707 km) 5925, gfs:29.75,-95.25 (16.671 km) 5926, gfs:30.00,-95.00 (27.258 km) 5926, gfs:29.50,-95.00 (30.354 km) 5925, gfs:30.00,-95.25 (31.0 km) 5926, gfs:29.50,-95.25 (33.77 km) 5926, gfs:29.75,-95.50 (40.758 km) 5927, gfs:30.00,-95.50 (48.382 km) 5927
noaa_gfs	pbl_height	gfs:29.75,-95.00 (7.707 km) 564.3, gfs:29.75,-95.25 (16.671 km) 790.8, gfs:30.00,-95.00 (27.258 km) 381.7, gfs:29.50,-95.00 (30.354 km) 478.2, gfs:30.00,-95.25 (31.0 km) 778.2, gfs:29.50,-95.25 (33.77 km) 429.8, gfs:29.75,-95.50 (40.758 km) 854.4, gfs:30.00,-95.50 (48.382 km) 785
noaa_gfs	precipitable_water	gfs:29.75,-95.00 (7.707 km) 36.89, gfs:29.75,-95.25 (16.671 km) 36.08, gfs:30.00,-95.00 (27.258 km) 37.17, gfs:29.50,-95.00 (30.354 km) 36.36, gfs:30.00,-95.25 (31.0 km) 36.71, gfs:29.50,-95.25 (33.77 km) 35.48, gfs:29.75,-95.50 (40.758 km) 35.04, gfs:30.00,-95.50 (48.382 km) 35.99
noaa_gfs	surface_pressure	gfs:29.75,-95.00 (7.707 km) 1012, gfs:29.75,-95.25 (16.671 km) 1011, gfs:30.00,-95.00 (27.258 km) 1010, gfs:29.50,-95.00 (30.354 km) 1013, gfs:30.00,-95.25 (31.0 km) 1010, gfs:29.50,-95.25 (33.77 km) 1012, gfs:29.75,-95.50 (40.758 km) 1010, gfs:30.00,-95.50 (48.382 km) 1008
noaa_gfs	t_850	gfs:29.75,-95.00 (7.707 km) 22.05, gfs:29.75,-95.25 (16.671 km) 22.03, gfs:30.00,-95.00 (27.258 km) 22.13, gfs:29.50,-95.00 (30.354 km) 22.11, gfs:30.00,-95.25 (31.0 km) 21.88, gfs:29.50,-95.25 (33.77 km) 22.26, gfs:29.75,-95.50 (40.758 km) 22.05, gfs:30.00,-95.50 (48.382 km) 22.09
noaa_gfs	u_10m	gfs:29.75,-95.00 (7.707 km) 0.678, gfs:29.75,-95.25 (16.671 km) 1.371, gfs:30.00,-95.00 (27.258 km) 0.7088, gfs:29.50,-95.00 (30.354 km) 1.516, gfs:30.00,-95.25 (31.0 km) 0.9855, gfs:29.50,-95.25 (33.77 km) 1.478, gfs:29.75,-95.50 (40.758 km) 1.818, gfs:30.00,-95.50 (48.382 km) 2.054
noaa_gfs	v_10m	gfs:29.75,-95.00 (7.707 km) 1.263, gfs:29.75,-95.25 (16.671 km) 1.315, gfs:30.00,-95.00 (27.258 km) 1.073, gfs:29.50,-95.00 (30.354 km) 1.8, gfs:30.00,-95.25 (31.0 km) 0.6752, gfs:29.50,-95.25 (33.77 km) 1.645, gfs:29.75,-95.50 (40.758 km) 1.054, gfs:30.00,-95.50 (48.382 km) 0.7177
openaq	ozone	2800 (0.0 km) 0.02268, 3214 (6.242 km) 0.02558, 339 (11.214 km) 0.02951, 332 (11.568 km) 0.02478, 2039 (13.761 km) 0.02326, 3378 (17.648 km) 0.01863, 5077791 (21.045 km) 0.01878, 2046 (21.111 km) 0.01974, +8 more
openaq	pm10	271 (11.568 km) 51.14, 15852419 (23.148 km) 37.44
openaq	pm25	268 (11.568 km) 17.42, 13787195 (13.589 km) 11.08, 15539682 (17.663 km) 9.662, 8967123 (17.705 km) 11.2, 8967479 (17.705 km) 10.83, 9489522 (18.242 km) 8.57, 9201803 (18.26 km) 9.943, 14157976 (18.483 km) 9.78, +65 more
openweather	cloud_cover	grid:east (3.216 km) 21.77, grid:center (28.161 km) 20.3, grid:north (37.319 km) 21.96, grid:south (37.957 km) 23.22
openweather	humidity	grid:east (3.216 km) 77.28, grid:center (28.161 km) 70.99, grid:north (37.319 km) 71.41, grid:south (37.957 km) 72.92
openweather	precipitation	grid:east (3.216 km) 0, grid:center (28.161 km) 0, grid:north (37.319 km) 0, grid:south (37.957 km) 0
openweather	pressure	grid:east (3.216 km) 1013, grid:center (28.161 km) 1013, grid:north (37.319 km) 1013, grid:south (37.957 km) 1013
openweather	temperature	grid:east (3.216 km) 30.16, grid:center (28.161 km) 31.14, grid:north (37.319 km) 30.55, grid:south (37.957 km) 30.25
openweather	wind_direction	grid:east (3.216 km) 214, grid:center (28.161 km) 227.2, grid:north (37.319 km) 204.7, grid:south (37.957 km) 216.3
openweather	wind_speed	grid:east (3.216 km) 2.463, grid:center (28.161 km) 1.848, grid:north (37.319 km) 1.912, grid:south (37.957 km) 2.329
purpleair	pm25	148709 (2.751 km) 10.71, 213701 (4.519 km) 9.855, 268171 (4.647 km) 9.021, 186271 (5.953 km) 10.89, 186277 (5.959 km) 12.21, 222815 (7.168 km) 9.338, 166427 (7.312 km) 13.43, 166441 (7.414 km) 11.09, +55 more
tceq	co	482011039 (11.598 km) 0.1131, 482011035 (17.662 km) 0.1533, 482011052 (30.401 km) 0.2641

Source	Metric	Values
tceq	no2	482011015 (0.618 km) 8.6, 482010026 (6.25 km) 6.727, 482011039 (11.598 km) 4.922, 482011035 (17.662 km) 7.032, 482011050 (21.051 km) 1.51, 482010416 (22.65 km) 5.125, 482010024 (28.341 km) 0.5348, 482011052 (30.401 km) 7.168, +4 more
tceq	so2	482011039 (11.598 km) 0.3475, 482011035 (17.662 km) -0.2833, 482010051 (41.337 km) 0.271





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The air quality in Houston, Texas is poor due to high levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

Local photochemical ozone production was likely enhanced near eastern Houston on 2026-07-20 because surface temperatures were very hot, cloud cover was minimal, humidity was relatively low for the area, and the ozone spike occurred during the afternoon photochemical peak.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

The openaq ozone network within 50 km had a mean concentration of 0.0208 ppm over the context window, which is well below the 0.072 ppm value observed at the anomaly site and time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

Elevated satellite-derived nitrogen dioxide and formaldehyde column densities, along with ground-based nitrogen dioxide readings, supplied the key chemical precursors required for peak ozone generation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

Urban and industrial precursor emissions likely contributed to the ozone spike near eastern Houston on 2026-07-20 because co-located nitrogen dioxide was elevated and satellite formaldehyde and nitrogen dioxide columns were higher than the previous day, while particulate indicators and carbon monoxide did not support a dominant wildfire-smoke cause.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

Elevated concentrations of nitrogen dioxide and formaldehyde provided the precursor reactants necessary for rapid photochemical ozone formation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

Surface weather observations recorded elevated temperatures exceeding 34°C, minimal cloud cover, and weak surface winds near 0 to 2 m/s during the period of the ozone anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

NO2 levels were also elevated, with a mean value of 11.5 ppb, suggesting that there was an increase in nitrogen dioxide emissions from nearby sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

Ground and satellite measurements indicated elevated concentrations of nitrogen dioxide and formaldehyde around the time of the ozone anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

The elevation of PM2.5 occurred at 2026-07-20T20:00:01+00:00, with a nearest distance of 27.812 km from the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Urban-industrial precursor emissions were a plausible contributor to the ozone spike, while wildfire smoke and a major sulfur-rich event were less supported, because coincident NO2 was elevated near the site and satellite NO2 and HCHO columns were higher than the previous day, whereas PM2.5 remained modest, CO was low, and satellite SO2 was not strongly elevated at the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

A strong wind pattern was observed on July 20th, potentially dispersing pollutants in the air and affecting air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The pollutant involved is particulate matter (PM2.5), which is elevated to 7.7 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Stagnant surface winds and weak regional advection reduced atmospheric dispersion, allowing photochemically generated ozone to accumulate near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

High ambient temperatures exceeding 34°C combined with clear sky conditions enhanced afternoon photochemical ozone production rates in Houston on July 20, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

Houston meteorology strongly favored rapid daytime ozone formation at the anomaly time because temperatures were very hot, cloud cover was very low, precipitation was absent, and surface winds were weak to calm near the site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The temperature in the area is unusually high, which may contribute to poor air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

There is a strong possibility of industrial or vehicular emissions contributing to the poor air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

Near-zero surface wind speeds and low wind vectors on July 20, 2026, restricted horizontal atmospheric dispersion, enabling localized ozone accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The air-quality anomaly in Houston, Texas on July 20th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The temperature was unusually high on July 20th, causing increased emissions from industrial and vehicular sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

The openaq ozone anomaly was detected by the zscore, stl, and isolation_forest methods at the same location and time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Weak dispersion and stagnation likely intensified the ozone peak near eastern Houston on 2026-07-20 because local surface winds near the anomaly were very light to calm while the boundary layer was deep enough to support daytime mixing and recirculation of aged precursor plumes.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

The openaq ozone measurement at 29.764, -95.078 was 0.072 ppm at 2026-07-20T20:00:00+00:00, while the expected value was 0.015873417721518984 ppm and the anomaly z-score was 5.817389994973134, classified as severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

Wildfire smoke or a major sulfur-rich plume was unlikely to be the dominant cause of the ozone anomaly because regional PM2.5 was not elevated, carbon monoxide was low, and sulfur dioxide evidence was limited despite some local afternoon SO2 presence.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

Hot, sunny, and dry afternoon conditions near eastern Houston on 2026-07-20 were strongly favorable for rapid photochemical ozone production, as surface temperature was about 34 to 36 C, cloud cover was about 6 percent, and relative humidity fell to roughly 47 to 57 percent near the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

Openaq particulate sensors situated 11.568 km from the anomaly site recorded maximum particulate levels of 61.0 ug/m3 for PM10 and 19.7 ug/m3 for PM2.5 concurrently at 2026-07-20T20:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

Weak local dispersion likely helped ozone and its precursors accumulate or age near the anomaly site on 2026-07-20, because the nearest ASOS observation reported calm wind at 19:55 UTC and GFS indicated a summertime afternoon boundary layer of about 1292 m with light 10 m winds near 1.1 m/s zonal and 0.1 m/s meridional flow.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

Houston precursor conditions supported local photochemical ozone production because satellite NO_2 and HCHO were elevated around the overpass time and nearby surface NO_2 was present at moderate levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

An openaq sensor located at coordinates (29.764, -95.078) in Houston registered a severe ozone spike reaching 0.072 ppm at 2026-07-20T20:00:00+00:00, significantly exceeding the expected baseline value of 0.016 ppm with a z-score of 5.82.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

The PM2.5 levels were higher than usual due to increased particulate matter emissions from nearby industrial activities or wildfires.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

An ozone anomaly reaching 0.072 ppm was observed in Houston at 20:00 UTC on July 20, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The PM2.5 levels were significantly higher than normal, with a mean value of 9.1181 ug/m3, indicating a high level of particulate matter in the air.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The location of the anomaly is in Houston, Texas, with no specific coordinates provided.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

Satellite observations from sentinel5p at 2026-07-20T19:49:35+00:00 near the anomaly site recorded elevated tropospheric columns of NO_2 at 5.83e-5 mol/m2 and HCHO at 0.000214 mol/m2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

High surface temperatures reaching up to 37.0°C combined with low cloud cover below 4 percent on July 20, 2026, created optimal photochemical conditions for rapid ozone formation in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.